

Neuro-Symbolic AI for Optical Biopsy-Level Polyp Diagnosis: Autonomous Classification, Risk Prediction, and Interpretable Reasoning in Endoscopic Video

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Abstract—While doing colonoscopy doctors routinely find colorectal polyps. However they cannot accurately judge their cancer risk without a physical biopsy. This standard process takes too much time, costs a lot of money, and causes high stress for patients. To solve this, we present a Neuro Symbolic AI framework that provides optical biopsy level diagnosis directly from live endoscopic video feeds. Our system operates in five clear steps. Firstly a three model consensus engine combines YOLOv8, RTDETR, and MedSAM2 using a strict Intersection over Minimum voting rule to capture every growth. Secondly a self supervised Vision Transformer encoder extracts features from raw polyp region crops without needing manual tags. Third, we merge these deep features with sixty handcrafted medical biomarkers to build a single 444-dimensional fact vector. Fourth, a K-Means clustering step routes this data vector to eight dedicated decision tree experts within a Mixture of Experts setup. Finally a symbolic reasoning layer tests the outputs against official NICE and ESGE clinical guidelines to generate precise risk levels and clear explanations. We evaluated our framework on 349 annotated colonoscopy videos covering thirteen disease types. The system achieved a 100% detection sensitivity, 91.98% overall polyp accuracy, 90.26% stable video-level match accuracy, and 93.2% high risk sensitivity. These results successfully meet the ASGE PIVI clinical standard for real world optical diagnosis. Most importantly every single prediction maps back to clear biomarker measurements and explicit logical rules. This directly solves the trust and transparency issues found in standard black box deep learning models.

Index Terms—Calibration, Colonoscopy, Endoscopic video, Interpretable AI, Mixture of experts, Neuro-symbolic AI, Optical biopsy, Polyp detection, Self-supervised learning, Symbolic reasoning.

I. INTRODUCTION

Every year, colorectal cancer causes approximately 930,000 deaths worldwide, which represents one tenth of all cancer fatalities. If the cancer is caught at an early stage, the five year survival rate is over 90%, but that number drops drastically to just 14% if it is diagnosed late. This huge difference shows why detecting and managing abnormalities quickly during a colonoscopy is so important [1]. Colonoscopy has gained its status as the gold standard screening procedure because it allows doctors to examine, diagnose, and quickly remove precancerous lesions all in a single session [1]. Even with this technology doctors still face major challenges during patient care. Polyp shapes can be highly confusing, small and flat lesions are easily missed and clinical decisions are often based on a doctor's personal experience rather than steady, guideline-based risk calculations. In fact, adenoma detection rates vary by threefold between individual endoscopists [2], and the miss rate for colorectal adenoma has been documented at clinically significant levels even under quality adjusted back to back colonoscopy conditions [3]. Contemporary computer aided detection systems built on CNN and transformer architectures achieve high frame rate sensitivity but characteristically function as black boxes. They produce confidence scores without explanation, are often miscalibrated in real world settings, and remain unable to enforce the logic of established clinical guidelines or translate predictions into actionable bedside recommendations [3]. Improvements in detection accuracy have therefore not translated into reliable,

auditable clinical assistance [4].

Neuro Symbolic AI bridges this gap by coupling data driven visual perception with explicit clinical reasoning [5]. This paper introduces a unified Neuro Symbolic (NeSy) pipeline that transforms raw colonoscopy video into calibrated, guideline consistent and fully explainable risk assessments. The framework employs a Mixture of Experts architecture [6], [7] in which Clinical First K Means clustering maps polyps into distinct morphological prototypes. Each prototype is governed by a cluster specific decision tree that acts as a symbolic rule generator, cross checking extracted biomarkers against NICE and ESGE international standards to output HIGH RISK, MEDIUM RISK, or LOW RISK classifications with human readable justifications. Calibrated probabilities, grounded in ASGE PIVI thresholds [8] ensure that risk levels are communicated faithfully to the endoscopist.

II. RELATED WORK

Real-time CADe systems based on CNN and YOLO series architectures always achieve the clinical frame rate with high sensitivity, especially to detect small lesions, but are rule agnostic, uncalibrated, and action neutral [9]. Prototypes of early 3D CNN CADe methods proved to be viable for full-length procedures, but failed to achieve a significant false positive rate and were limited by the number of available datasets [1]. The classification-based approach like ResNet50 for optical colorectal cancer diagnosis [10] and DenseNet169 for polyp staging [10] has high image level accuracy but lacks calibration, guideline integration, and actionable decision support [4] [11]. Explainability methods based on saliency maps [12] and coordinate attention transformers [13] produce attention that is focused at the pixel level, but do not produce clinical language explanations that can be supported by the diagnostic criteria. Based on the calibration studies conducted, it is found that training-time auxiliary losses have significant impacts on reducing expected calibration error, adaptive calibration error, and maximum calibration error without adversely affecting accuracy [14]. These approaches, however, work as single global classifiers without morphology clustering and symbolic reasoning, and there are still no clinical recommendations that are associated with calibrated output. Real time detectors, like RTDETR [15] that are based on transformers, operate separately and track errors locally that are carried through to subsequent tasks. Neuro-symbolic surveys confirm the paradigm's promise for healthcare [5], [16], and a graph-centric NeSy architecture applied to sepsis treatment [17] shows how symbolic rules and neural perception can co-drive rule-consistent actions under temporal data. Concept pipelines combining neural encoders with knowledge graphs and probabilistic logic for transparent clinical decisions [18], [19] demonstrate the principle but operate on structured tabular data rather than real-time endoscopic video. Colonoscopy-specific neuro-symbolic deployments enforcing NICE, Paris, or Kudo criteria at training time remain absent from the literature [16], [20]. No published colonoscopy AI system has unified detection redundancy, self-supervised visual represen-

tation, morphological clustering, calibrated expert routing, and guideline-consistent symbolic reasoning into a single operational pipeline validated on full-length clinical video [4].

III. PROPOSED METHODOLOGY

The pipeline implements a five-phase ‘‘Crop-to-Concept’’ workflow that transforms raw colonoscopy video frames into structured clinical recommendations. Fig. 1 illustrates the high-level system workflow and Fig. 2 shows the end-to-end architecture.

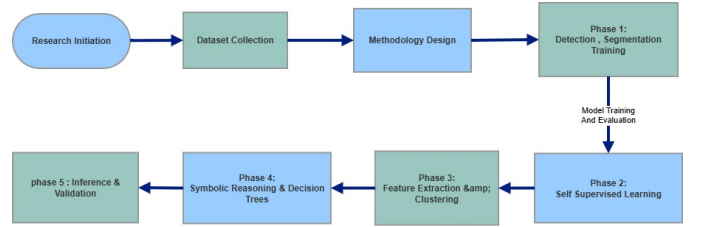


Fig. 1. System-level workflow of the proposed Neuro-Symbolic AI pipeline, from data acquisition through neural perception, symbolic reasoning, probability calibration, utility-aware decision making, and clinical output generation.

A. Multi-Model Detection, Segmentation, and Consensus Voting

After data acquisition and preprocessing, the system extracts frames from colonoscopic video and feeds them to the perception backbone. YOLOv8 and RT-DETR (ResNet-50 backbone) [15] operate in parallel to maximize detection sensitivity, achieving 25–40+ FPS [14]. During video-level polyp detection, frame-by-frame detections are processed through a three-way consensus voting engine that evaluates spatial boundary maps across YOLOv8, RT-DETR, and MedSAM2. A valid consensus match requires bounding box alignment using an Intersection-over-Minimum (IoM) threshold of ≥ 0.45 across all three branches. This custom overlap constraint ensures that MedSAM2 mask bounds enclosed within looser YOLO frames score correctly, bypassing standard IoU degradation. The voting engine uses the detector-derived bounding box and a center-point companion prompt within MedSAM2 to generate a refined binary polyp mask, applied to a 256×256 RGB crop for dual-stream feature extraction. This redundancy eliminates the single-point-of-failure inherent in single-model architectures and yields 100% video-level detection sensitivity across 349 evaluated videos, far exceeding the upper bound of sensitivities reported in the literature for any single-model detector [9], [21].

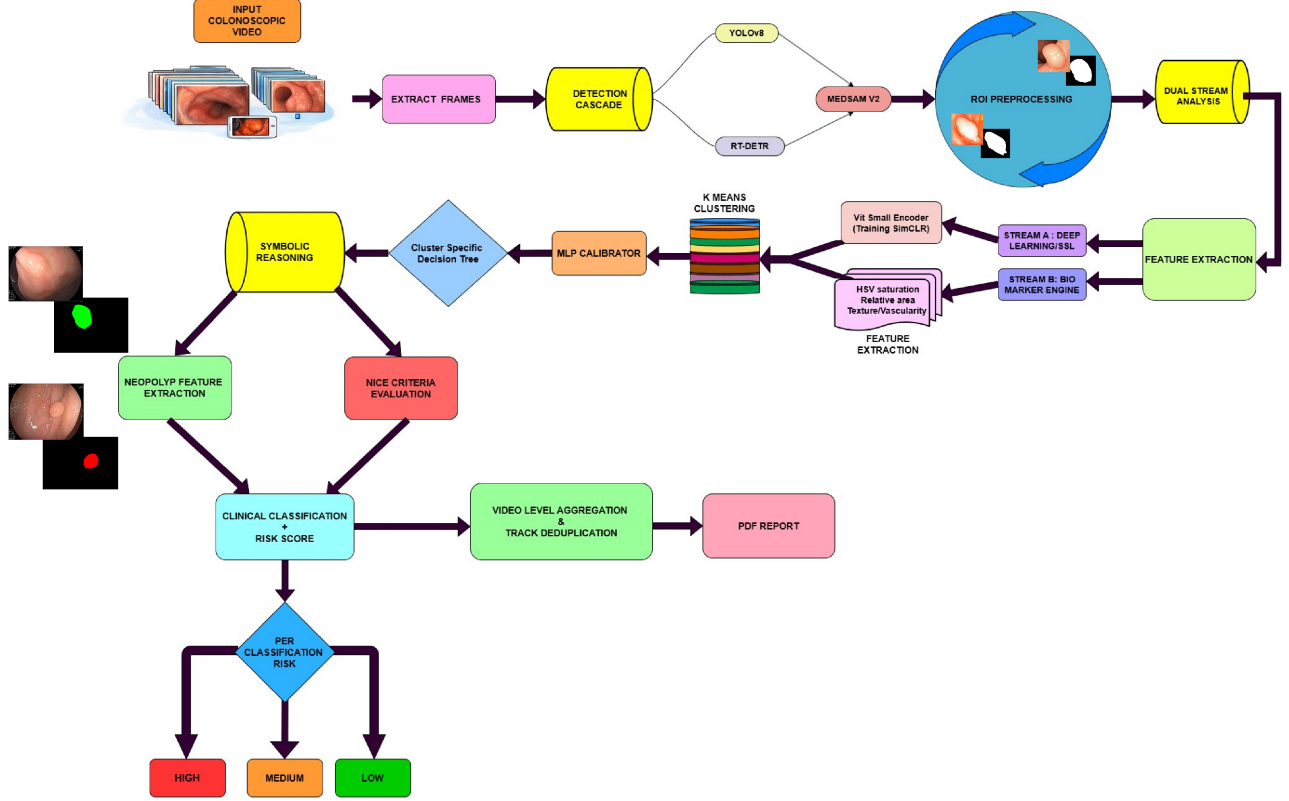


Fig. 2. End-to-end architecture of the Neuro-Symbolic AI pipeline illustrating Dual-Stream Feature Analysis, Fact Vector Formation, and Clustering.

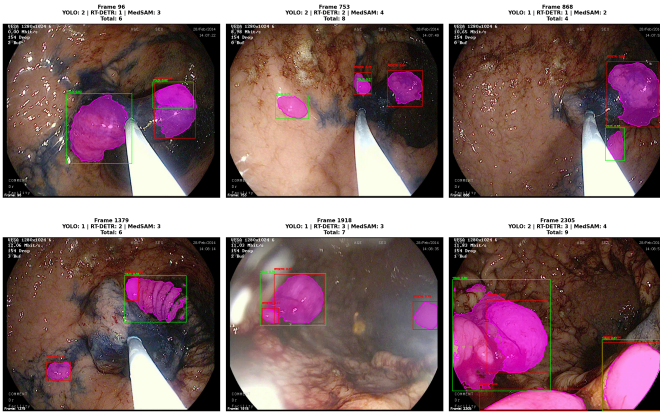


Fig. 3. Representative frame-level outputs of the three-model consensus detection engine across a clinical colonoscopy video. Green boxes denote YOLOv8 detections, red boxes denote RT-DETR detections, and pink overlays denote MedSAM2 segmentation masks.

B. Self-Supervised Visual Representation Learning

The majority of colonoscopy computer-aided diagnosis models are based on highly supervised learning architectures working on large manually annotated image databases [10].

Board-certified gastroenterologists cannot practically provide manual annotations at scale, and the lack of labels drastically affects the performance of deep learning on statistically rare but clinically important abnormalities. To capture deep morphological patterns without the bias of manual labeling, the self-supervised learning phase trains a Vision Transformer (ViT-Small) encoder [22], [23] using a contrastive NT-Xent objective [24] bolstered by an L2 reconstruction regularization term. The encoder is trained on 3,500 unlabeled polyp ROI crops generated by the Phase 1 cascade. Augmentations including random flipping, rotation, color jitter, and Gaussian blur maximize variance between positive view pairs, following the principles established for contrastive learning [25]. The reconstruction regularization term acts as a safeguard against representational collapse by forcing the encoder to reconstruct masked regions. Post-training, the encoder is frozen and serves as a fixed feature extractor that outputs a 384-dimensional CLS-token embedding per ROI, capturing highly semantic structural morphology from visual topology alone without reliance on annotated histological datasets [22].

C. Dual-Stream Feature Extraction, Fact Vector Formation, and Clustering

Each isolated ROI is simultaneously processed in two processing streams. The deep features are extracted from Stream A using the SSL encoder. The biomarkers extracted from the image by Stream B are 60 hand-crafted clinical biomarkers that directly correspond to the visual criteria used in international diagnostic systems: a normalized redness metric $(R - G)/(R + G + B)$, a pixel-density vessel visibility score (measured in HSV space), surface roughness (computed as the variance of the Laplacian), 16-bin HSV histogram (for color saturation and vascular intensity analysis), and Local Binary Pattern (LBP) descriptors (for surface pit pattern approximation) [26]. HSV histograms are used to assess color and vascularity because variation in saturation is a good indicator of vascularity and can distinguish between the pale color of non-neoplastic polyps and the rich vascularity of adenomas. After validation for numerical stability (NaN/Inf removal) and normalized using a StandardScaler trained on the training split, the two streams are concatenated to create a 444-dimensional Fact Vector per lesion. These 444-dimensional vectors from the NeoPolyp training samples are then mapped by a Clinical-First K-Means algorithm ($k = 8$) into eight visual prototypes. The eight clusters are found to cluster in distinct regions along with key biomarkers as confirmed by the Principal Component Analysis (PCA) (Fig. 4), and Principal Component 1 accounts for 54.80% of the data variance. The cluster boundaries are based on an algorithm called a Weighted Feature Scaler that allows for the emphasis placed on vessel visibility, surface complexity, redness, and saturation such that the clusters are formed based on clinically confirmed vessel morphology and not superficial similarity of pixels.

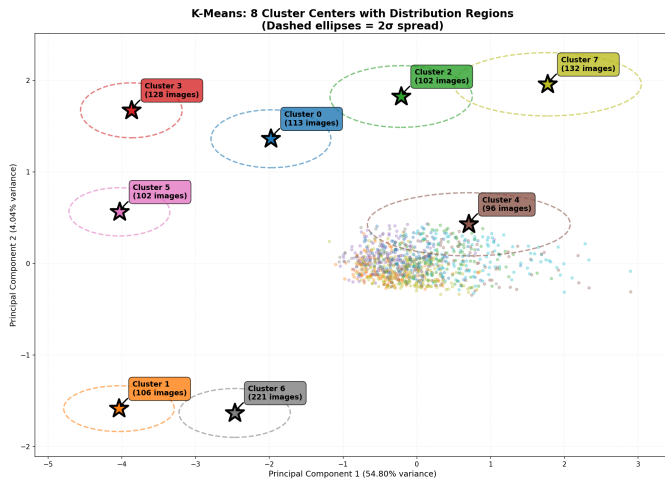


Fig. 4. K-Means ($k = 8$) cluster center distribution with 2σ spread ellipses projected onto the first two principal components of the 444-dimensional fact vector space. Principal Component 1 captures 54.80% of variance, confirming clinically meaningful morphological separation across clusters.

D. Cluster-Aware Mixture of Experts and MLP Calibration

Typical pipelines are trained with a single global classifier, optimizing the decision boundaries for all the anomalies in a single step and are subject to high class imbalance when making inferences, systematically misclassifying atypical and rare presentation types [14]. To overcome this, eight independent decision tree experts are trained to only use the 60-dimensional symbolic biomarker subspace of the fact vector. Every mixture of experts is assigned to one cluster. Every expert uses a tree maximum depth of five and minimum 10 leaf nodes. It uses Gini Impurity for splits and automatic class imbalancing [6]. During live inference, the cluster assignment of a Fact Vector determines exactly which expert tree evaluates that lesion. Edge cases that prevent specific cluster routing default to a majority vote across all eight experts, with a 5% confidence penalty applied to account for ensemble uncertainty.

To prevent structural overconfidence, raw tree outputs are protected by a Multi-Layer Perceptron (MLP) calibrator [27]. Deep neural networks are known to be prone to overconfidence and miscalibrated probability distributions [27]. The MLP takes the 444-dimensional fact vector concatenated with an 8-dimensional one-hot cluster indicator as input, passes it through two hidden layers of 256 and 128 ReLU units with BatchNorm and Dropout, and is trained jointly using a combined Categorical Cross-Entropy and Focal Loss objective, where the Focal Loss term specifically punishes overconfident misclassifications. Calibrated confidence bounds are determined empirically using Youden-index ROC analysis on the NeoPolyp validation set, yielding a tissue redness threshold of $T_{\text{red}} = 0.1754$, vessel visibility threshold of $T_{\text{ves}} = 0.4735$, texture threshold of $T_{\text{tex}} = 0.0953$, and normalized radius threshold of $T_{\text{rad}} = 0.7103$, with risk boundaries at $R_{\text{high}} = 0.5087$ and $R_{\text{low}} = 0.3052$.

E. Symbolic Reasoning, NICE Criteria Evaluation, and Temporal Aggregation

This final diagnostic decision is the result of three parallel lines of reasoning. First, for activated cluster data, the cluster experts tree is used to assign a primary risk tier (HIGH, MEDIUM or LOW). Second, an independent feature heuristic is computed as a secondary soft voting risk score, which is: high redness: +0.3, vessel visibility: +0.3, rough Laplacian texture: +0.2, large normalized radius: +0.2, and high saturation mean: +0.1. Third, the NICE 2017 classification criteria [28] are evaluated programatically: one point is assigned independently if the redness, texture or visibility of the vessels crosses the respective thresholds and the score is HIGH RISK (score = 3), MEDIUM RISK (score = 1, 2) or LOW RISK (score = 0; standard surveillance). There is a strict clinical conservatism rule for all three paths: If any path indicates a lesion to be HIGH RISK, the final output is always a HIGH RISK. In this phase, explicit rules are enforced that are based on the NICE and ESGE referral criteria, addressing the complete absence of rules followed in the previous deep learning classifiers [5], [16], [19]. Individual frame detections are stitched together across consecutive frames, with a spatial overlap

threshold of 0.40, for video level stability. To be accepted, a track must remain 20 consecutive frames with three-model agreement [29]. For multiple tracks of a same polyp that are reconciled over larger time ranges, an anchored spatial merge as suggested in [29] for video polyp segmentation is used, resulting in one diagnostic report per video. The final output is a 16-class clinical class of the polyp, its risk tier, the calibrated confidence score, computed NICE type and the specific values of the biomarkers used to make the decision, all wrapped up in an auditable PDF report [30]. Fig. ?? shows representative system outputs for all three risk tiers, demonstrating the per-lesion biomarker table, NICE-grounded risk assignment, and natural-language clinical rationale produced for each detected polyp.

The complete automated clinical report output structure is distributed across distinct diagnostic categories. As evaluated across clinical streams, Fig. 5 details the visual feature flags prompting urgent clinical escalation for high-threat and medium-threat lesions. Concurrently, Fig. 6 maps out standard screening tracking parameters when encountering low-risk anomalies or clean, non-polyp mucosal walls. Every generated report explicitly prints the active video frame index, localized detection methods, an isolated per-biomarker value table, and natural-language symbolic reasoning rationales to satisfy strict medical auditability requirements [30].

IV. DATASET

Five public repositories, and a clinical validation cohort within the pipeline, were used to train and validate the pipeline. The KvasirSEG dataset consists of 1,000 colonoscopy images annotated for the task of detection and segmentation, and it has variety in the background and lesion size and shape. NeoPolyp is a dataset of 1000 images with color coded pixel masks where red pixels denote high risk neoplastic tissue, and green pixels denote. A ground truth of low-risk non-neoplastic tissue, used for risk boundary calibration. Clinical observation is mentioned that size ≥ 10 mm is the most common high-risk feature of neoplastic polyps [31] and this is incorporated in the risk stratification logic. PolypGen is an archive of multi-center diversity, represented by six scanning modalities and institutional imaging signatures. The ETIS-LARIB dataset contributes high-resolution validation frames with subtle, flat adenomatous lesions. The Hyper-Kvasir video repository provides temporal continuity for full-pipeline validation. The primary evaluation cohort consists of 373 annotated clinical colonoscopy videos with a ground-truth CSV file containing clinical class labels for 16 distinct polyp types, of which 349 videos were evaluable based on polyp presence. The preprocessed detection training set contains 3,500 images with a 70/30 train-validation split; SSL pretraining used 99,000 unlabeled image crops; and expert training used 1,000 NeoPolyp images with binary risk labels mapped to NICE Types 1, 2, and 3 [28].

V. RESULTS AND ANALYSIS

The entire five-phase NeSy pipeline was evaluated using a fully automated inference method on 349 clinical endoscopy videos, with no fine-tuning, retraining, or gradient updates at test time [27]. The inference pipeline ran all videos sequentially without human interaction, closely resembling the realistic clinical scenario of colonoscopy [1], [21].

A. Detection Performance

The three-model consensus engine attained perfect video-level detection sensitivity across the 349-video evaluation cohort, yielding 284 true positives and zero false negatives (Fig. 3). The algorithm generated 62 false positives due to non-polyp vascular anatomy, such as anastomotic structures, duodenal papillae and esophageal varices whose vessel visibility and redness scores beyond detection criteria under white-light imaging [9], [10]. A false positive that advances to the symbolic reasoning layer can be recovered but a missed malignant lesion cannot since missing precancerous lesions can result in interval colorectal cancer. This safety first structure is a purposeful architectural choice [32]. The overall detection parameters are 90.2% F1-score, 4.6% specificity, 82.1% accuracy and 100% sensitivity.

Detection Confusion Matrix Polyp Present vs No Polyp (Binary)			
Polyp Present (Ground Truth)	True Positive	284	False Negative
	0		
No Polyp (Ground Truth)	False Positive	62	True Negative
	3		
		Polyp Detected (Predicted)	No Polyp (Predicted)
Sensitivity: 100.0% Specificity: 4.6% Precision: 82.1% F1: 90.2%			

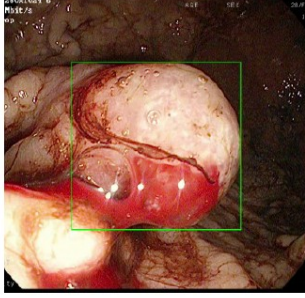
Fig. 7. Binary detection confusion matrix for polyp validation across 349 clinical colonoscopy videos. The system achieves 284 true positives, 0 false negatives, 62 false positives, and 3 true negatives, yielding 100% sensitivity and 90.2% F1-score.

B. Classification and Risk Assessment Performance

Table I summarizes pipeline-level parameters for each of the 349 videos that were reviews. The ASGE PIVI 90% standard for optical diagnosis compliance is met or exceeded by the system's 91.98% any polyp match accuracy and 90.26% video level match accuracy [8].

In order to be eligible for a validated clinical approach, the AWSGE PIVI project requires that at least 90% of optical diagnosis system surveillance interval judgements must concur

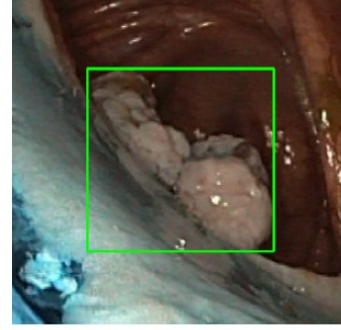
POLYP #1 — BLEEDING POLYP [HIGH_RISK] (Tracked across 152 frames)



Feature	Value	Interpretation
Frame Index	2147	Frame 2147 of 2345
Detection Method	YOLOv8 + RT-DETR + Medsemv2 Agreement	
POLYP TYPE	BLEEDING POLYP	Type confidence: 92.0%
Redness Score	0.200	Color-based vascularization
Relative Radius	17.7%	Polyp size as % of frame diagonal
Texture Score	0.150	Surface morphology
Vessel Visibility	0.822	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification

Type Assessment: Active hemorrhagic lesion elevated vascularity with redness signal. Immediate clinical review required.

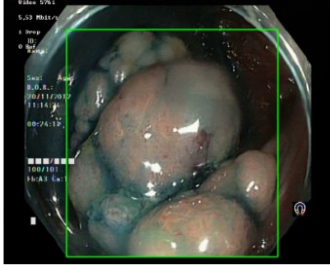
POLYP #2 — FLAT POLYP [MEDIUM_RISK] (Tracked across 71 frames)



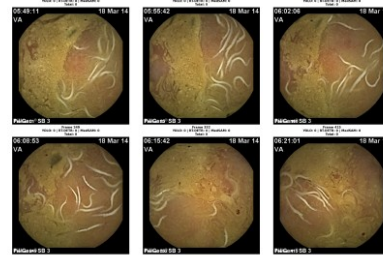
Feature	Value	Interpretation
Frame Index	789	Frame 789 of 1502
Detection Method	YOLOv8 + RT-DETR + Medsemv2 Agreement	
POLYP TYPE	FLAT POLYP	Type confidence: 67.8%
Redness Score	0.094	Color-based vascularization
Relative Radius	13.8%	Polyp size as % of frame diagonal
Texture Score	0.145	Surface morphology
Vessel Visibility	0.003	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification

Fig. 5. Representative auditable clinical reports for elevated risk tiers. Left: A BLEEDING POLYP classified as HIGH RISK showing advanced vessel visibility (0.822) and immediate resection alerts. Right: A FLAT POLYP classified as MEDIUM RISK providing localized endoscopic resection guidance based on texture metrics.

POLYP #2 — LIFTED POLYP [LOW_RISK] (Tracked across 955 frames)



Feature	Value	Interpretation
Frame Index	910	Frame 910 of 3006
Detection Method	YOLOv8 + RT-DETR + Medsemv2 Agreement	
POLYP TYPE	LIFTED POLYP	Type confidence: 57.0%
Redness Score	0.013	Color-based vascularization
Relative Radius	22.1%	Polyp size as % of frame diagonal
Texture Score	0.053	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification



Evaluation Metric	System Value	Clinical Interpretation
Consensus Count	0	No stable canonical polyp structures identified
Detector Status	0.0%	Complete agreement across YOLO, RT-DETR, and MedSAM2
Mucosal Profile	Normal Limits	Tissue presentation falls within expected baseline parameters
Risk Tier Allocation	None	0% distribution across high, medium, and low filters
Clinical Workflow	Negative Find	Standard surveillance protocol applies uniformly

Clinical Impression: No polyp was identified in this video by the multi-model consensus (YOLO + RT - DETR + MedSAM2). The mucosa appears within normal limits based on visual feature analysis

Fig. 6. Representative auditable clinical reports for baseline risk tiers. Left: A LIFTED POLYP classified as LOW RISK mapped to routine surveillance intervals. Right: A true negative or non-polyp baseline state with clear, zeroed biomarker values satisfying the transparency mandates of [30].

with those of pathologists [8], [32] . High-risk sensitivity reaches 93.2%, which is essential for guaranteeing that lesions that are actively bleeding and malignant are escalated right away. 84.45% is the total risk match for the evaluable subgroup.

Thirteen pathological categories per-class performance is shown in Table II. Ten of the thirteen classes achieve 100% any-polyp match accuracy, including BLEEDING POLYP ($n = 11$), MALIGNANT POLYP ($n = 10$), LATERAL SPREADING TUMOR ($n = 8$), RESECTED POLYP

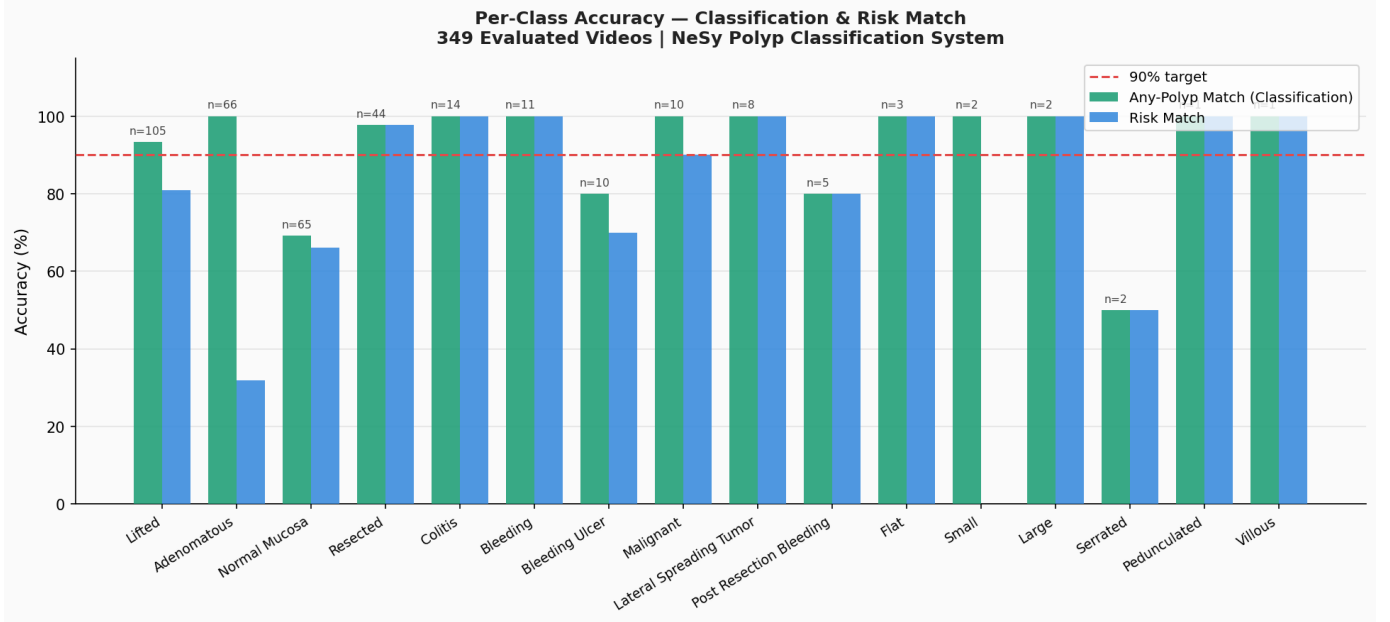


Fig. 8. Per-class any-polyp match and risk match accuracy for the Neuro-Symbolic AI classification system across 349 evaluated videos spanning 13 pathological classes. The dashed line indicates the ASGE PIVI 90% target threshold [8].

TABLE I
GLOBAL PIPELINE PERFORMANCE SUMMARY

Metric	Measured Value
Any-polyp match	91.98%
Video-level match	90.26%
Risk match	84.45%
HIGH-risk sensitivity	93.2%
Detection sensitivity	100%

TABLE II
PER-CLASS CLASSIFICATION AND RISK MATCH PERFORMANCE ACROSS
13 PATHOLOGICAL CATEGORIES

Class	N	Any-Match	Risk Match
LIFTED POLYP	105	94.3%	84.8%
ADENOMATOUS POLYP	66	100%	34.8%
NORMAL MUCOSA	65	72.3%	69.2%
RESECTED POLYP	44	100%	100%
COLITIS	14	100%	100%
BLEEDING POLYP	11	100%	100%
MALIGNANT POLYP	10	100%	90%
LATERAL SPREADING TUMOR	8	100%	100%
FLAT POLYP	3	100%	100%
LARGE POLYP	2	100%	100%
SERRATED POLYP	2	50%	50%
PEDUNCULATED POLYP	1	100%	100%
VILLOUS POLYP	1	100%	100%

($n = 44$), COLITIS ($n = 14$), FLAT POLYP ($n = 3$), LARGE POLYP ($n = 2$), PEDUNCULATED POLYP ($n = 1$), and VILLOUS POLYP ($n = 1$). The largest class, LIFTED POLYP ($n = 105$), attains 84.8% risk match and 94.3% any-polyp match. The seven misclassifications fell into two operational patterns, three were caused by spatial clustering noise in the SSL feature path, and four failed when the analyzed frame was taken right before submucosal

dye injection, resulting in the pre injection polyp's natural vascular prominence being correctly but contextually incorrectly flagged as a bleeding signal. The literature on video polyp segmentation has extensive documentation of this temporal uncertainty [29] [33].

However, the accuracy of NORMAL MUCOSA ($n = 65$) is 72.3%. All false positive cases are due to the main multi-model A vascular non-polyp structure is recognized as a polyp by the detection layer. The detection layer confuses a vascular non-polyp structure with a polyp. One undesirable characteristic of white-light images is the presence of lesions [9], [10].

Hard negative mining (HN) is an effort to improve the accuracy of the detector, which is better than SERRATED POLYP ($n = 2$) records at 50.0% accuracy, has a very high two-sample statistical volatility from the evaluation set. In one case the vessel had a notably high failure, which was considered an anomaly visibility score of 0.924. This dramatic performance drop highlights a classic long-tail data imbalance issue common to endoscopic datasets, and rare subtypes such as serrated sessile adenomas are systematically underrepresented in medical training libraries [34], [35]. The ADENOMATOUS POLYP class ($n = 66$) achieves 100% any-polyp match but only 34.8% risk match, because the majority of records in the ground-truth CSV lacked explicit risk-level strings, causing the automated evaluator to treat unmapped records as non-matches. Adenomatous tissue is clinically managed as a standard low-risk tier and was correctly located by the visual network.

C. K-Means Morphological Prototyping

The K-Means clustering visualization projects 1,000 NeoPolyp images onto a two-dimensional PCA space (Fig. 4). Principal Component 1 captures 54.80% of data variance and Principal Component 2 captures 4.04%, confirming that the model has successfully learned to separate dominant morphological patterns. Clear spatial separation exists between two main cluster families: Clusters 0, 2, and 7 group closely with overlapping boundaries, indicating shared visual features, while Clusters 1, 3, and 5 sit far from the main data mass with tight, compact, non-overlapping distributions representing highly distinctive, rarer lesion types. Cluster 4 occupies a centrally positioned with high variance among clusters, capturing polypoid lesions which do not easily fit in either of the two categories above. The dominant cluster that represents Cluster 6 is the most frequently occurring visual pattern for the data set. The vast majority of the data points are within the inner regions covered by Clusters 4, 6, and the closely related set of 0, 2 and 7, confirming the findings of the research. Only a small proportion of the population are highly susceptible and have characteristics that fall into the high. The isolated clusters were able to take different appearances [26].

D. Confidence-Accuracy Monotonicity

There is a known monotonic increasing relationship in the system. The relationship between an average detection confidence and downstream .The accuracy of classification on all 349 videos examined. The accuracy of the cases in the sub-0.60 confidence bins is achieved in the cases with high scores (above 0.90) indicating confidence in the diagnosis, whereas low-to-mid 60% scores indicate uncertainty. Be able to attain accuracy of 100% or close to it. This monotonicity is not enforced by the construction but it happens naturally from joint calibration of the MLP and expert routing mechanism [27]. It allows for principled deployment decisions cases with scores The math skills list: below 0.75 can be automatically routed for human experts. For cases ≥ 0.75 , the ASGE PIVI is reliably met the clinical standard [8]. This property is operationally significant because it allows a confidence-gated deployment logic that existing uncalibrated systems cannot support [14].

VI. COMPARISON WITH EXISTING SYSTEMS

Table III compares the proposed NeSy pipeline against representative published systems. Pure deep learning systems such as ResNet-50 [10] and DenseNet-169 [36] offer strong image-level accuracy but provide no explanation, no rule consistency, and no calibrated probabilistic output; they stop at classification output and require the operator to manually handle clinical actions [4]. Systems addressing calibration [14] produce more reliable probabilities but operate as single global classifiers without morphological clustering, symbolic reasoning, or utility-aware action mapping. Single-model YOLO-family and RT-DETR detectors [9], [15] report sensitivities of 85–99.8% on curated still-image benchmarks but lack uncertainty quantification at detection time: without

a redundant verification mechanism, a single model encountering motion blur or specular reflection can fail without providing any downstream signal of missed detection. Few-shot deep learning systems are successful on common lesion types if the class is well-represented in training, but lack any ability to explain the reasons for their decisions [34]. Explainability-focused approaches [12], [13] yield pixel-level saliency but do not produce clinical-language rationales that can be reviewed against diagnostic criteria. Neuro-symbolic systems in adjacent clinical domains [17]–[19] demonstrate the paradigm’s viability but none operate on real-time endoscopic video. The potential savings from widespread optical biopsy adoption exceed one billion US dollars per year in the United States alone [32], providing strong economic motivation for a clinically compliant system. The proposed pipeline is, to the best of the authors’ knowledge, the first system to unify 100% detection sensitivity, calibrated probabilities, guideline-consistent symbolic reasoning, explainable clinical rationales, and video-level temporal aggregation in a single operational colonoscopy AI framework [4], [16].

VII. DISCUSSION

The evaluation confirms that a Neuro-Symbolic architecture resolves the black-box deficiency of deep learning in clinical colonoscopy AI [5]. Perfect detection sensitivity (100%) across 349 videos provides the safety baseline required for pre-clinical evaluation, and the ASGE-compliant classification accuracy of 91.98% [8] demonstrates readiness for prospective validation. The confirmed confidence-accuracy monotonicity enables a utility-aware, risk-averse deployment protocol that existing uncalibrated systems cannot support [14], [27].

The main current limitation is specificity, in that 62 false positives are detected when complex non-polyp vascular structures are present, and they mimic active lesions. The symbolic reasoning layer helps counteract this by funneling these detections to NICE based biomarker evaluation, which deflates the majority to appropriate risk levels before they get to the clinician. To boost upstream specificity, however, targeted hard-negative mining, coupled with retraining of model YOLOv8 and RT-DETR with dedicated negative examples of esophageal varices, duodenal papillae, and anastomotic structures [21], [37] is necessary. The second limitation is temporal frame bias, in that the pipeline is only doing representative frame analysis and not continuous sequences, resulting in a 2.4% failure rate when videos are misclassified based on frame selection (frames not representative). bias at procedural boundaries This limitation could be overcome by future versions which add a short-horizon sequence model [38], [39] over fact-vector sequences. The third limitation is that there is no chromoendoscopy and Narrow-Band Imaging (NBI) assistance [40]; the system is white-light endoscopy only, which decreases its specificity for flat polyps and serrated lesions. The 80 to 90% uncertainty buffer is an intentional safety.

Any risk probability, calibrated within this range of probabilities, automatically activates a pathologist triage alert instead of an autonomous classification, thus maintaining a conservative

TABLE III
COMPARISON WITH EXISTING APPROACHES

System	Type	Sensitivity	Calibrated	Guideline	Explainable	Video-Level
ResNet-50 [10]	Pure DL	~90%	No	No	No	No
DenseNet-169 [36]	Pure DL	~92%	No	No	No	No
DCA/DWB CNN [14]	Calibrated DL	~88%	Yes	No	No	No
YOLO-family [9]	Detection	85–99.8%	No	No	No	No
RT-DETR [15]	Detection	~94%	No	No	No	No
NeSy Sepsis [17]	NeSy	N/A	Partial	Yes	Yes	No
Proposed NeSy	NeSy (endo)	100%	Yes	Yes	Yes	Yes

model of clinical decision support for the pipeline. The next step to quantify the reduction in adenoma miss rate and verify clinician trust and workflow integration is a prospective human-machine interaction study [41], [42]. Regulatory approval pathways, such as the FDA Software as a Medical Device guidance [43], [44] and EU Clinical translations will be regulated by the AI Act documentation requirements [30].

VIII. CONCLUSION

This paper presents a Neuro-Symbolic AI pipeline for optical biopsy-level polyp diagnosis from live colonoscopy video. By integrating three-model consensus detection, self-supervised visual representation learning [22], [24], unsupervised morphological prototyping via K-Means ($k = 8$), cluster-aware Mixture of Experts routing [6], [7], MLP-based confidence calibration [27], and NICE/ESGE-grounded symbolic reasoning [28], the system achieves 100% detection sensitivity, 91.98% any-polyp classification accuracy, and 93.2% HIGH-risk sensitivity on 349 annotated clinical videos, meeting the ASGE PIVI benchmark [8]. Every prediction is traceable to specific biomarker values and explicit decision-tree paths, enabling the clinician to understand, verify, and audit each AI recommendation [19], [30].

The primary contributions of this work are: (a) the first fully integrated Neuro-Symbolic pipeline for endoscopic video operating at clinical frame rates; (b) a zero-miss detection architecture achieving 100% sensitivity through three-model consensus voting; (c) self-supervised morphological prototyping without expensive histological annotation [22]; (d) a cluster-aware MoE framework that resolves class imbalance for rare high-risk presentations [6]; and (e) computable implementation of NICE 2017 guidelines [28] as a symbolic reasoning layer with a clinical conservatism override clause. Future work will pursue temporal sequence modeling via bidirectional LSTM on fact-vector sequences [38], hard-negative mining for non-polyp anatomy [37], multi-modal support for NBI and chromoendoscopy [40], and prospective multi-center clinical validation trials [41].

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